Nontraditional Aviation Operations for Wildfire Response

Fiscal Year 2025 SBIR Phase I Solicitation 86

Scope Description:

In the United States, wildfires are becoming increasingly severe and costly in terms of acreage burned, property damaged, and most importantly, lives lost. Wildfire frequency and intensity is escalating, inducing budgetary, personnel, and equipment challenges. Furthermore, California and other western states have been facing persistentdrought conditions and much hotter temperatures, which are fueling wildfire intensity and duration. These alarming trends have made it urgent to better predict, mitigate, and manage wildland fires. NASA's history of contributions to wildfire and other disaster management efforts includes remote sensing, instrumentation, mapping, data fusion, and prediction. More recently, NASA ARMD has been investigating capabilities to help manage wildfire suppression and mitigation efforts through technologies for coordination of airspace operations for wildfire management. NASA ARMD has recently made a significant contribution to enable widespread use of small, unmanned aircraft systems (sUAS) by developing air traffic management capabilities for low-altitude unmanned vehicle operations, called UAS Traffic Management (UTM). This work is being adapted to safely and efficiently integrate larger vehicles and operations with existing operations and mission types. NASA recognizes the value these capabilities could provide when applied to the aerial wildfire management domain. Current applications of aviation to wildfire management include deployment of smoke jumpers to a fire; transport of firefighters, equipment, and supplies; fire retardant or water drop; reconnaissance of fire locations and fire behavior; and supervision of air tactical operations.

Current challenges of aerial wildfire management include the following:

- Existing airspace management techniques are manual and cannot accommodate new aircraft types suitable for wildfire response operations (e.g., unmanned aircraft).
- Aerial firefighting is limited to acceptable visual conditions (no night operations).
- Monitoring and remote-sensing missions are intermittent, flown outside of active firefighting or available periodically from satellite assets.
- There is a lack of reliable, resilient, and secure data communications for quick information dissemination to support effective decision making.

NASA is seeking technologies to:

- Provide strategic planning capabilities to collect, process, and disseminate information that enables persistent monitoring of wildland fire conditions (e.g., satellites, conventional aircraft, and UAS).
- Provide strategic planning and tracking capabilities to enable the most effective use of ground crews, ground equipment, and aircraft during operations (e.g., both at a single incident and across multiple incidents).
- Provide strategic planning capabilities that support multi-mission planning to support efficient mission assignments to support concurrent operations (e.g., air attack and search and rescue).
- Provide an extension to the UTM network that considers the unique needs and characteristics of wildfire disaster situations (e.g., non-connected environments) and the response to combat them.
- Increase the throughput of available communications, reduce the latency of data transfer, provide interoperability with existing communication solutions, and provide a reliable network for the use of UAS, other aviation assets, and emergency responders on the ground.
- Provide a mobile position, navigation, and timing solution to support automated operations (e.g. automated precision water drops) in Global Positioning System (GPS) degraded environments (e.g., mountainous canyons).
- Provide wildland fire prediction, airspace coordination, and resource tracking for a common operating picture for situational awareness that supports various stakeholders in the incident command structure (e.g., incident commander, air tactical group supervisor, aircraft dispatch, UAS pilot, etc.).
- Ensure the highest safety and efficiency of operations.

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Proposers wanting to focus on services or technologies to coordinate airborne operations across a wildfire area should submit their proposal to the current subtopic scope.

Proposals focused on the following will be rejected for this subtopic:

- Technologies that help autonomous or piloted flight in areas with degraded visibility.
- Technologies that enable single-pilot multi-ship operations.

- Technologies that support unmanned logistic operations such as moving supplies to a different area.
- Technologies that support wildfire suppression and management missions.

Expected TRL or TRL Range at completion of the Project: 1 to 5

Primary Technology Taxonomy:

• Level 1: TX 16 Air Traffic Management and Range Tracking Systems

• Level 2: TX 16.3 Traffic Management Concepts

Desired Deliverables of Phase I and Phase II:

• Research

• Analysis

• Prototype

• Software

Desired Deliverables Description:

NASAs intent is to select proposals that have the potential to move a critical technology or concept beyond Phase II SBIR funding and transition it to Phase III, where a NASA aeronautics program, another government agency, or a commercial entity in the aeronautics sector can fund further maturation as needed, leading to actual usage in future airspace operations.

The Phase I outcome should establish the scientific, technical, and commercial feasibility of the proposed innovation in fulfillment of NASA objectives and broader aviation community needs. Phase I should demonstrate advancement of a specific technology or technique, supported by analytical and/or experimental studies that are documented in final report.

Phase II efforts should yield:

1. Models supported with experimental data,

2. Software related to a model that was developed,

3. A material system or prototype tool, or

4. Modeling tools for incorporation in software, etc. that can be infused into a NASA project or that lead to commercialization of the technology.

Consequently, Phase II efforts are strengthened when they include a partnership with a potential end-user of the technology.

Phase I award recipients must be thinking about commercialization and which organizations will be able to use the technology following a Phase II effort. It is necessary to take that into account, rather than just focusing on developing technology without putting a strong effort into developing a commercial partner or setting the effort up for continued funding by teaming with an organization post-Phase II.

State of the Art and Critical Gaps:

The current state of the art for coordination of aerial firefighting is a manual process that must be coordinated across multiple entities, often bringing multiple aerial assets to the wildfire fighting environment. Advanced tools and techniques are required to address the following gaps:

- Existing airspace management process is very manual and slow.
- Awareness of aircraft operations is conducted by visual monitoring and radio communication.
- Unmanned systems are not easily integrated into aerial fire suppression operations.
- Operations are limited by visibility and no operations are conducted at night, when fires often die back.
- Surveillance images are captured and disseminated only every 4 hours.
- Intermittent communication can delay effective response.
- Conditions can rapidly change, requiring timely information for effective decision making.
- Decision makers for emergency response are overloaded with data.
- Information requirements differ for various roles within the disaster response.
- Tools and data are often spread across numerous applications.

Relevance / Science Traceability:

Due to climate change, wildfires are becoming increasingly more frequent and severe. Fire seasons are longer, lasting 6 to 8 months; in some cases, fire season is year-round. The 2020 fire season was the worst in recorded history, burning over 4 million acres of land, destroying more than 8,500 structures, and killing more than 30 people. The economic impact of these fires is in the hundreds of billions of dollars and results in lasting societal impact. The annual cost of fire suppression has soared from roughly $425 million per year in 1999 to $1.6 billion in 2019. On June 30, 2021, President Biden and Vice President Harris met with governors from western states, Cabinet officials, and private-sector partners to discuss specific actions the public and private sectors are each taking to strengthen prevention, preparedness, mitigation, and response efforts to protect communities across our country from wildfires and their devastating impacts. The President directed several actions, in close coordination with state and local governments and the private sector, to ensure the federal government can most effectively protect public safety and deliver assistance to our people in times of urgent need.

References:

1. Airspace Operations and Safety Program (AOSP): https://www.nasa.gov/aeroresearch/programs/aosp

2. NASA Aeronautics Research Mission Directorate (ARMD) Strategic Implementation